

Learning Analytics Dashboard Using OULAD

Project Report:

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Tools Used

- Python
- Pandas
- Power BI
- DAX
- GitHub

1. Executive Summary:

Educational institutions generate large volumes of student performance and engagement data. However, transforming this information into actionable insights remains a significant challenge.

This project developed an end-to-end Learning Analytics solution using the Open University Learning Analytics Dataset (OULAD). The objective was to analyze learner demographics, assessment performance, and digital engagement patterns to identify factors associated with academic success.

The project involved:

- Data acquisition
- Data quality assessment
- Data cleaning
- Dimensional data modeling
- Business intelligence development
- Interactive dashboard creation

The final Power BI dashboard enables stakeholders to monitor student performance, compare learner segments, and evaluate the relationship between engagement and academic outcomes.

2. Business Problem:

Educational institutions seek to answer several important questions:

- Which learner groups achieve the strongest academic outcomes?
- Does prior educational background influence performance?
- How does online engagement affect final results?
- Which course modules have the highest success rates?
- Can engagement metrics be used as early indicators of academic risk?

Answering these questions can help universities improve student support programs, increase retention, and enhance learning outcomes.

3. Dataset Description:

Dataset Source:

Open University Learning Analytics Dataset (OULAD): <https://analyse.kmi.open.ac.uk/open-dataset>

The dataset contains information on student demographics, assessment results, registration activity, academic outcomes, and Virtual Learning Environment (VLE) interactions.

Source Files Reviewed:

`courses.csv`

Contains:

- Course module information
- Presentation periods
- Course duration

`assessments.csv`

Contains:

- Assessment identifiers
- Assessment type
- Assessment weighting
- Assessment schedules

`studentAssessment.csv`

Contains:

- Student assessment submissions
- Scores
- Submission timelines

`studentInfo.csv`

Contains:

- Student demographics
- Education level
- Geographic region
- Age group
- Disability status
- Final academic result

studentRegistration.csv

Contains:

- Registration dates
- Withdrawal dates

studentVle.csv

Contains:

- Student engagement activity
- Daily interaction records
- Clickstream data

vle.csv

Contains:

- Learning resource metadata
- Activity categories

4. Data Quality Assessment:

A structured data profiling process was performed on each source file.

The assessment included:

- Dataset dimensions
- Data types
- Missing values
- Duplicate records
- Unique value analysis
- Descriptive statistics
- Business rule validation

Key Findings

studentAssessment

Metric	Value
Records	173,912
Missing Scores	173
Unique Students	23,369
Unique Assessments	188
Negative Submission Dates	2,057

studentInfo

Metric	Value
Records	32,593
Missing IMD Band	1,111
Duplicate Rows	0

studentRegistration

Metric	Value
Missing Registration Dates	45
Missing Withdrawal Dates	22,521

studentVle

Metric	Value
Records	10,655,280
Duplicate Interaction Rows	787,170
Missing Values	0

5. Data Cleaning Process:

Several cleaning and transformation activities were performed to improve analytical quality.

Missing Value Handling:

studentAssessment

Missing assessment scores were retained and flagged to avoid introducing artificial values into performance metrics.

studentInfo

Missing IMD Band values were assigned an “Unknown” category.

studentRegistration

Missing withdrawal dates were interpreted as learners remaining enrolled through course completion.

Duplicate Handling:

The studentVle dataset contained 787,170 duplicate interaction records.

Instead of removing these records, interaction activity was aggregated to preserve engagement volume while reducing dataset size.

Data Validation:

Validation checks were performed to ensure:

- Assessment identifiers matched assessment definitions.
- Student identifiers remained consistent across datasets.
- Registration and engagement records aligned correctly with module presentations.
- Categorical values remained standardized.

6. Data Modeling:

A dimensional modeling approach was adopted to support analytical reporting.

Analytical Tables Created:

Dimension Tables

- dim_studentInfo
- dim_courses
- dim_assessments
- dim_student_registration

Fact Tables

- fact_student_assessment
- fact_student_engagement

Additional VLE resource metadata was prepared during data modeling for future analytical expansion.

7. Performance Optimization:

Challenge:

The original studentVle dataset contained over 10.6 million records, resulting in slow refresh and report performance.

Solution:

Student interaction data was aggregated at:

- Student Level
- Module Level
- Presentation Level
- Date Level

The following metrics were created:

- Total Clicks
- Activity Count

Result:

Dataset	Rows
Original studentVle	10.6 Million
Aggregated Engagement Table	1.8 Million

This reduced dataset size significantly while preserving analytical value.

8. Power BI Semantic Model:

The final dashboard implementation utilized the following core analytical tables:

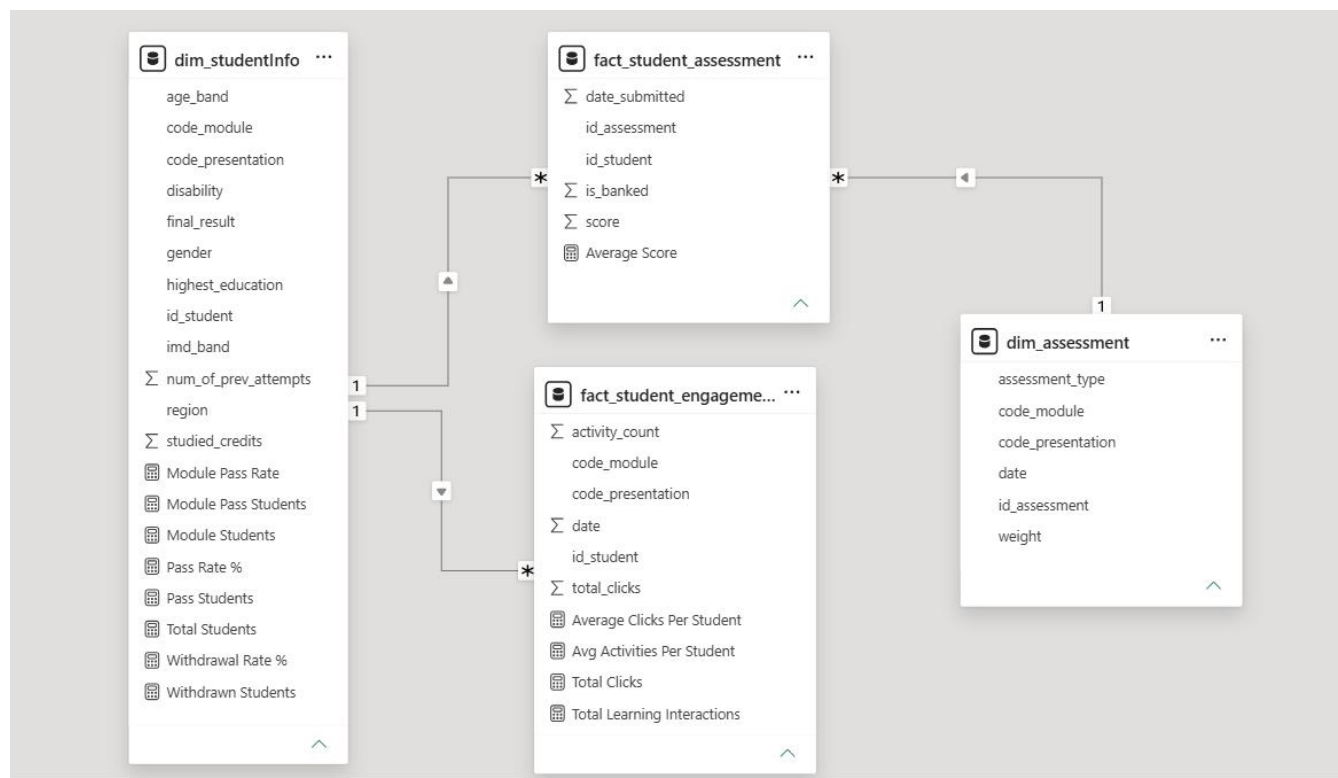
Dimension Table

- dim_studentInfo
- dim_assessment

Fact Tables

- fact_student_assessment
- fact_student_engagement

This simplified model provided optimal dashboard performance while maintaining analytical depth.



9. DAX Measures:

Several measures were created to support business analysis.

Average Assessment Score: Measures average learner performance across assessments.

Total Students: Counts unique students.

Pass Students:

Counts learners achieving:

- Pass
- Distinction

Pass Rate %: Calculates overall student success rate.

Formula: $\text{Pass Students} \div \text{Total Students}$

Average Clicks Per Student: Measures average digital engagement.

Average Activities Per Student: Measures average interaction frequency.

Total Clicks: Measures total VLE engagement volume.

10. Dashboard Design:

The dashboard was designed as a three-page analytical solution.

Page 1: Executive Overview

Purpose:

Provide a high-level summary of student outcomes and engagement.

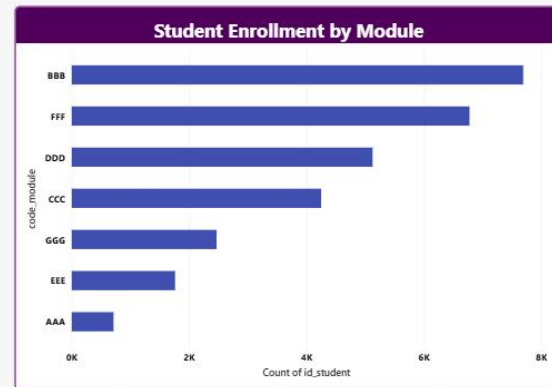
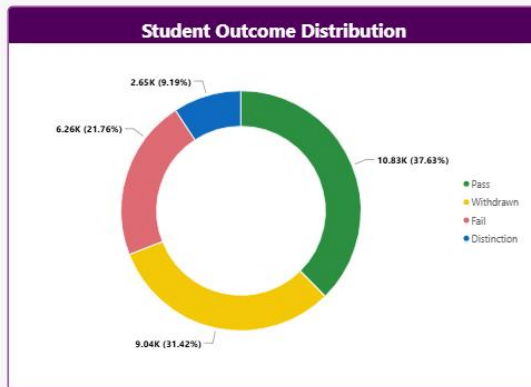
KPIs:

- Total Students
- Average Assessment Score
- Pass Rate
- Total Clicks

Business Value:

Provides leadership with a quick view of academic performance and learner engagement.

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Page 2: Student Performance Analysis

Business Questions:

- Does educational background influence performance?
- Which age groups perform best?
- Which regions have stronger outcomes?
- Which modules have the highest pass rates?

Visuals:

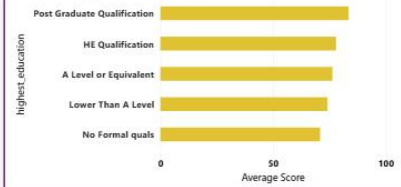
- Average Score by Education Level
- Average Score by Age Band
- Student Distribution by Region
- Pass Rate by Module
- Final Result Distribution

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Average Assessment Score

75.80

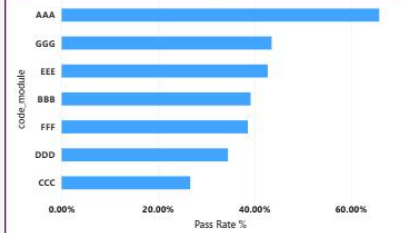
Average Score by Education Level



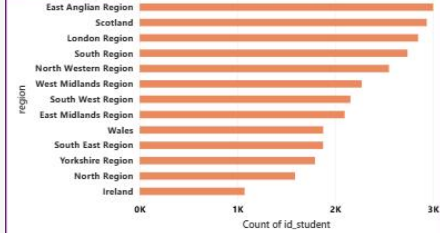
Average Score by Age Band



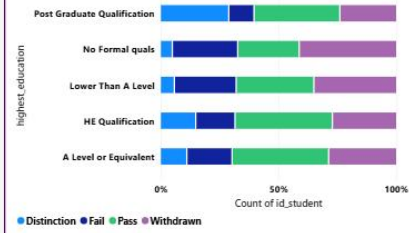
Pass Rate by Module



Student Count by Region



Final Result Distribution by Education Level



Page 3: Engagement & Learning Behavior Analysis

Business Questions:

- Do engaged learners achieve stronger outcomes?
- Which modules drive the most activity?
- How does engagement vary across demographic groups?

Visuals:

- Average Clicks by Final Result
- Average Activities by Final Result
- Total Clicks by Module
- Average Clicks by Education Level
- Average Clicks by Age Band

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Average Clicks Per Student

1.52K

Average Activities Per Student

5.89

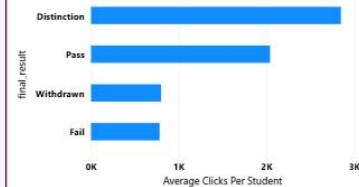
Total Learning Interactions

11M

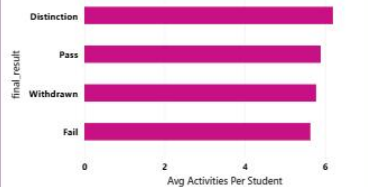
Total Clicks

40M

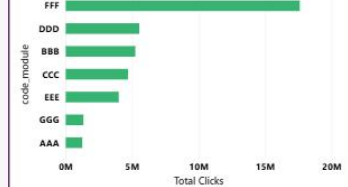
Average Clicks by Final Result



Average Activities by Final Result



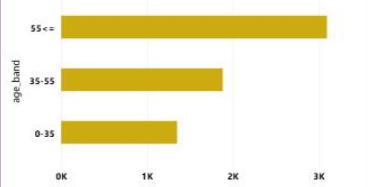
Total Clicks by Module



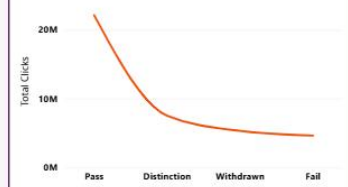
Average Clicks by Education Level



Average Clicks by Age Band



Engagement vs Academic Outcome



11. Key Insights:

Academic Performance:

- Student outcomes vary considerably across course modules.
- Educational background shows measurable influence on academic performance.
- Certain modules exhibit significantly lower success rates.

Learner Engagement:

- Students with stronger engagement generally achieve better academic outcomes.
- Learners who passed or achieved distinction demonstrated substantially higher interaction levels.
- Engagement metrics provide useful indicators of learner success.

Retention Opportunities:

- Registration and engagement data can be leveraged to identify at-risk learners.
- Declining engagement patterns may provide early warning signals before assessment failure occurs.

12. Challenges Encountered:

During development, several challenges were addressed:

- Managing a 10.6 million-row engagement dataset.
- Handling duplicate interaction records.
- Designing relationships between multiple educational datasets.
- Optimizing Power BI performance.
- Creating meaningful business metrics from educational data.

These challenges provided valuable experience in large-scale analytical project development.

13. Future Enhancements:

Potential future improvements include:

- Student retention analysis using registration data.
- Withdrawal prediction modeling.
- Cohort analysis across presentation periods.
- Machine learning models for student success prediction.
- Resource-level engagement analysis using VLE metadata.

14. Conclusion:

This project demonstrates a complete analytics workflow from raw educational data to business-ready dashboards.

The solution successfully transformed large-scale learner data into actionable insights that help explain student performance, engagement behavior, and academic outcomes.

The project showcases practical skills in data cleaning, dimensional modeling, business intelligence development, and data-driven storytelling using real-world educational data.